North South University

Department of Electrical and Computer Engineering

SocietyCare

Community Problem Reporting and Resolution System

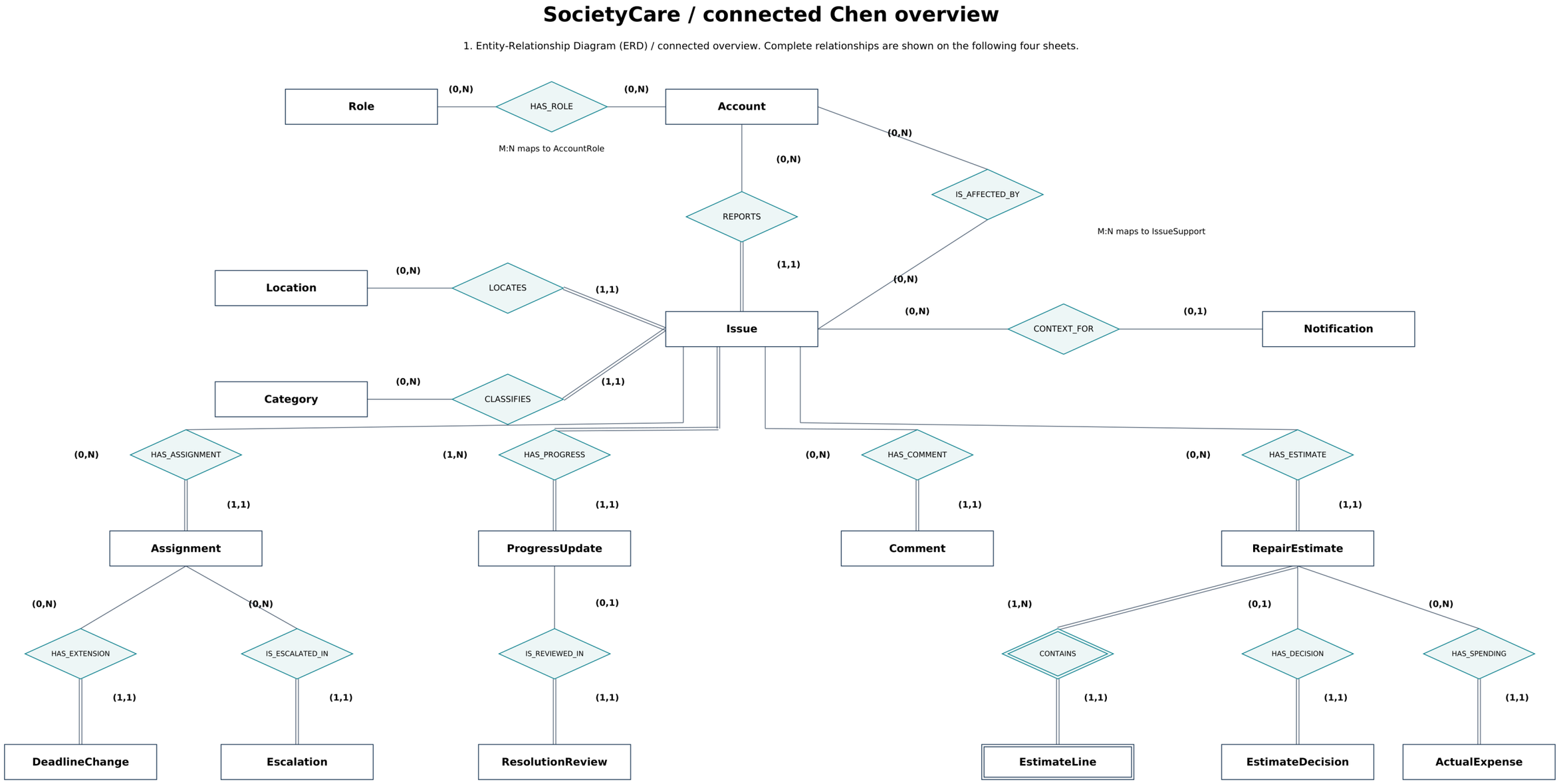
Entity Relationship Diagram and Database Schema Design

Course code	CSE311L
Course title	Database Systems Lab
Lab instructor	Md. Amanullah Shah
Section	3
Group number	3
Submission date	05/11/2026

Group Members

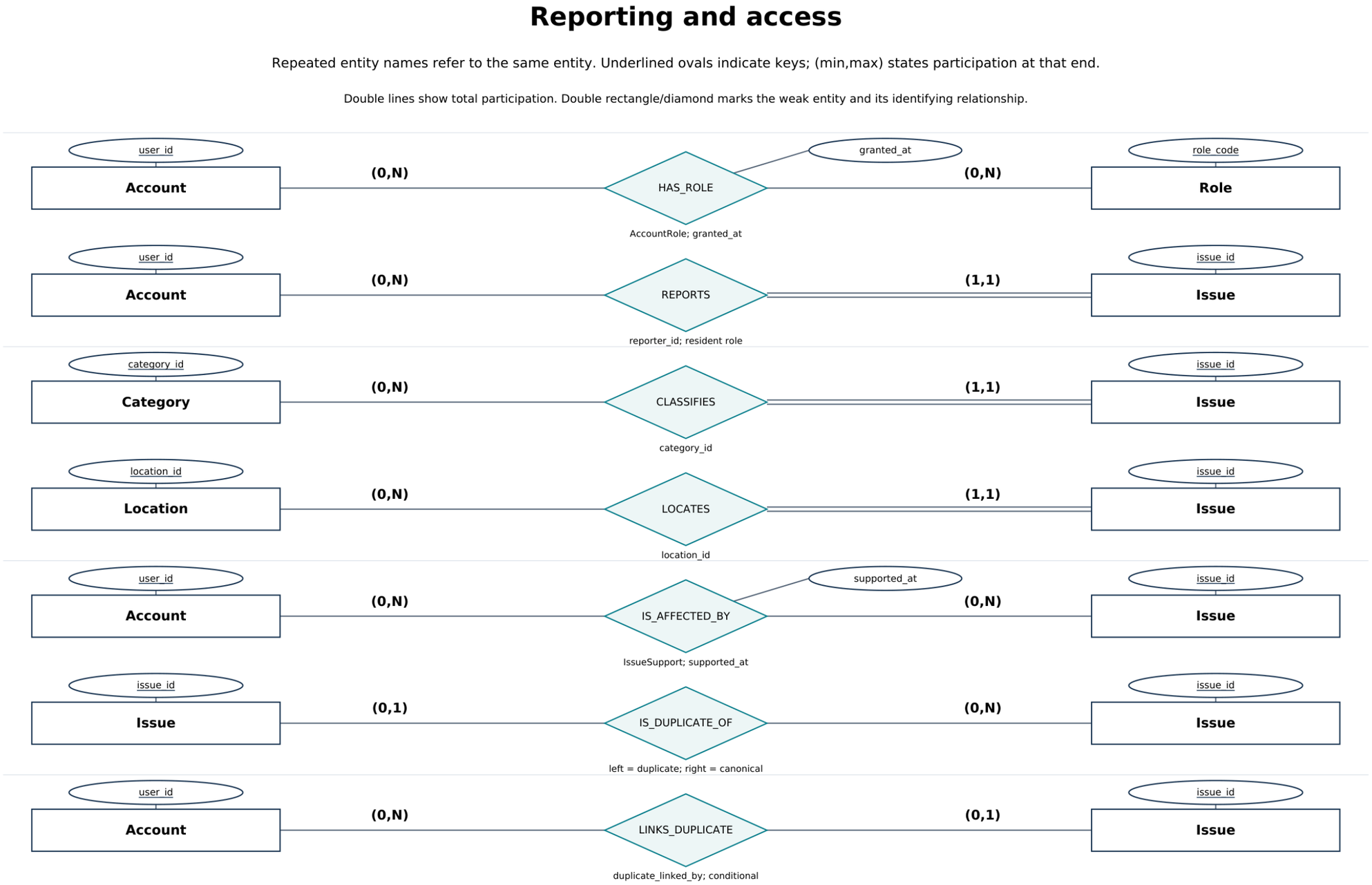
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1 Entity Relationship Diagram

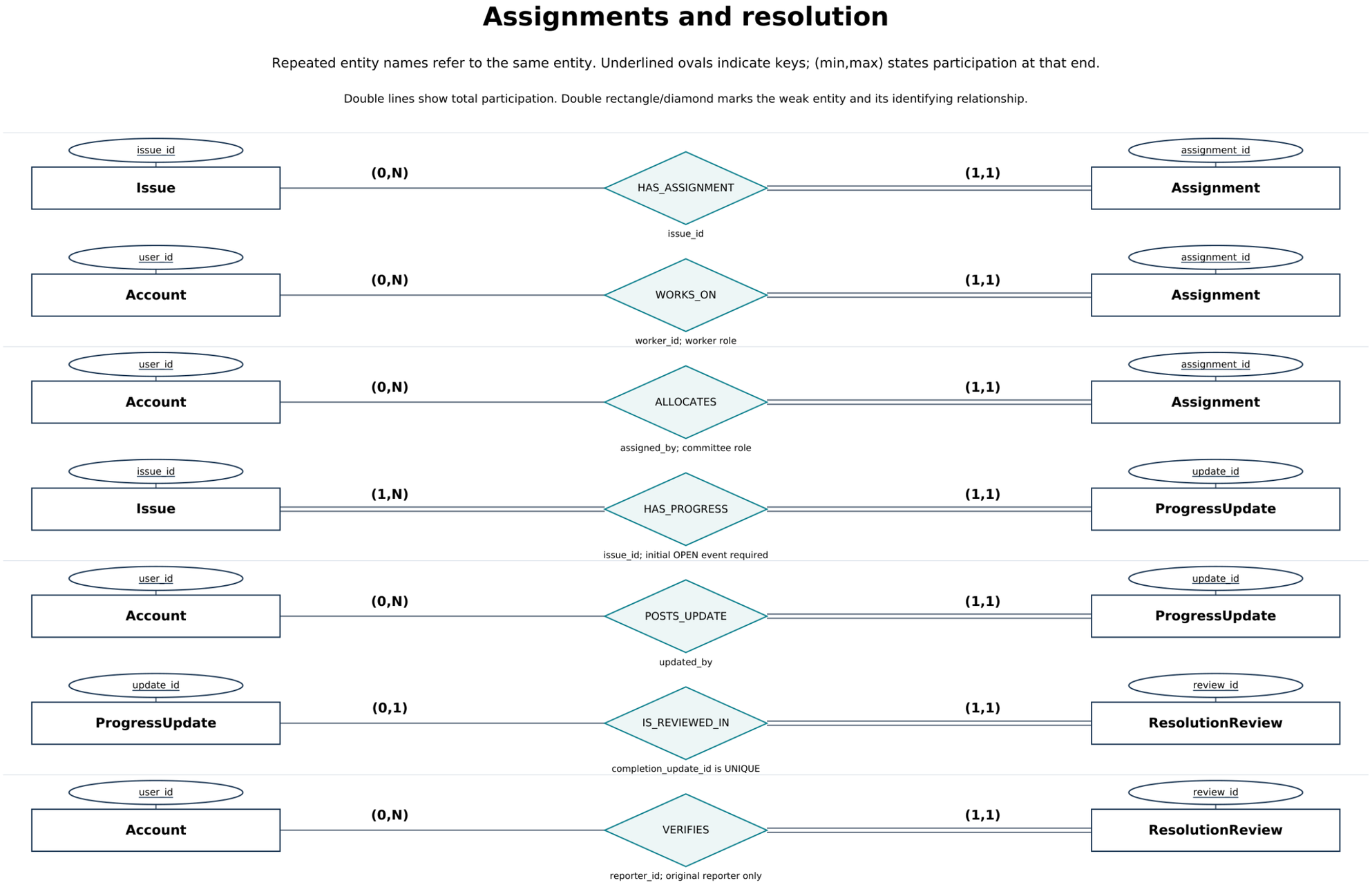


Chen notation: rectangles = entities; diamonds = relationships; double rectangle/diamond = weak entity/identification; double line = total participation.

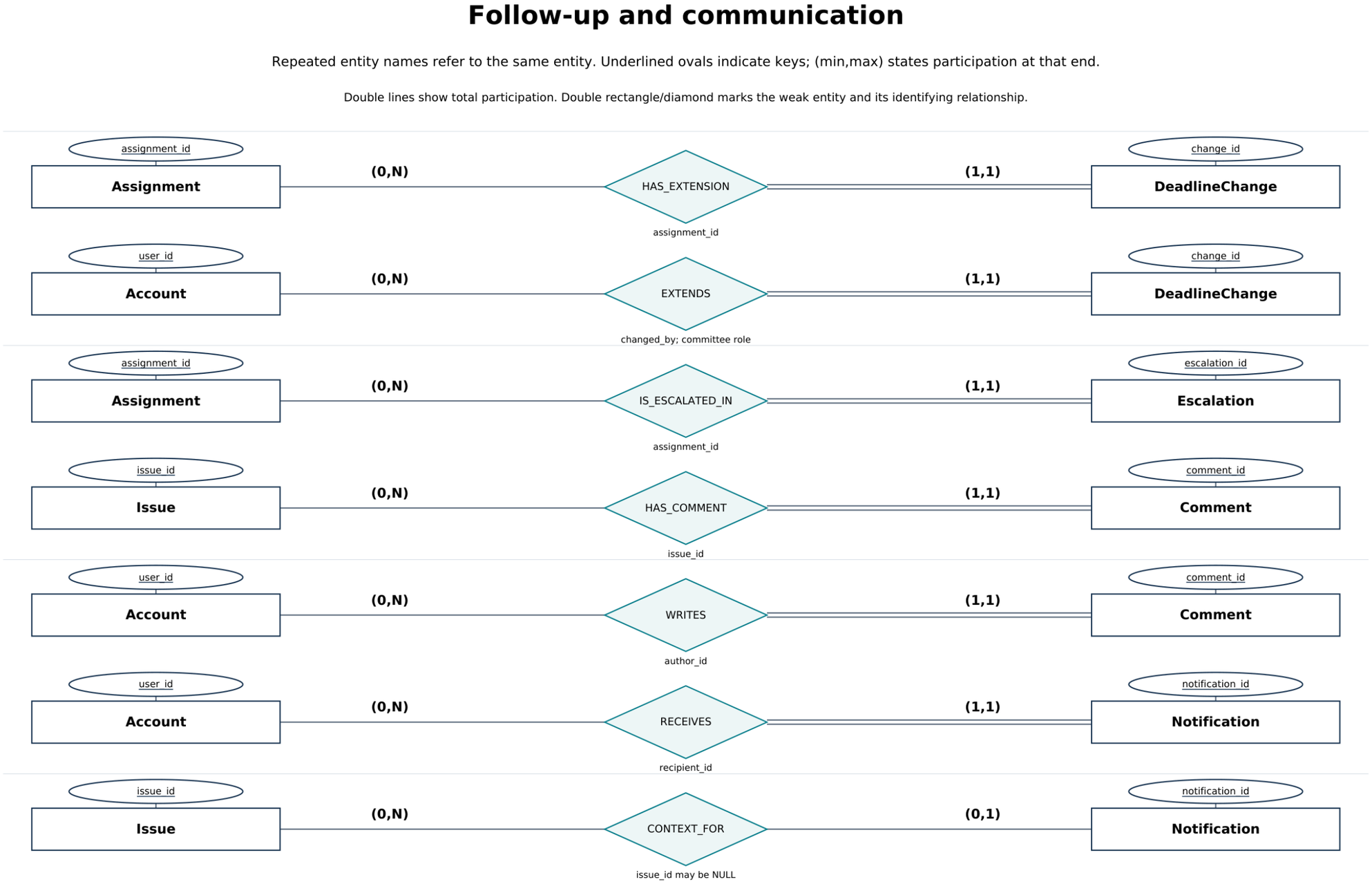
1 ERD Reporting and Access



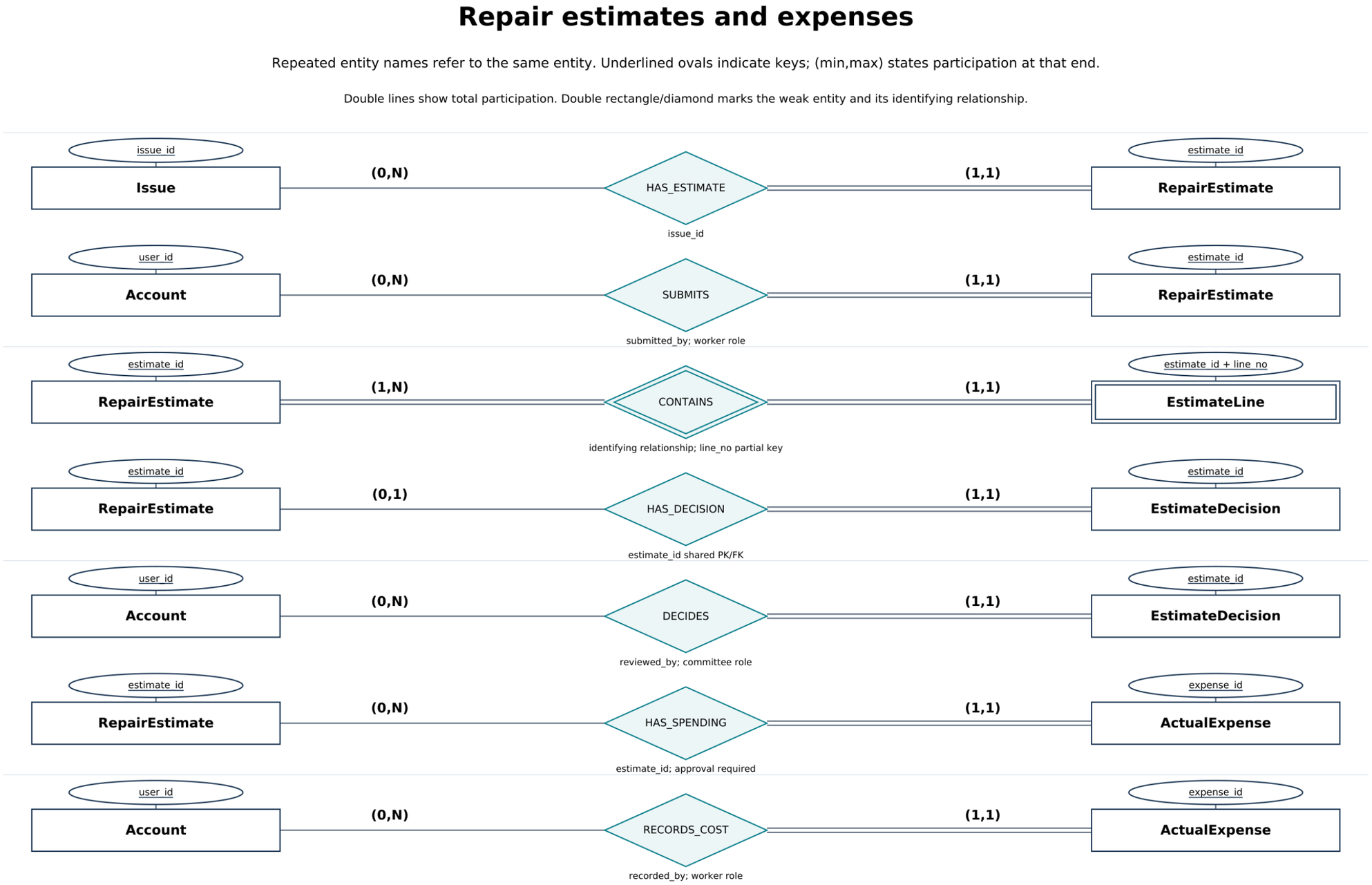
1 ERD Assignments and Resolution



1 ERD Follow Up and Communication



1 ERD Repair Estimates and Expenses



2 List of Entities and Attributes

Underlined = primary key; *italic* = foreign key; *both styles* = both keys.
Multiple underlined attributes together form a composite primary key.

Account
user_id
full_name
phone
password_hash
house_flat
created_at
is_active

Role
role_code
role_name

AccountRole
user_id
role_code
granted_at

Location
location_id
block_name
road_name
landmark

Category
category_id
category_name
description

Issue
issue_id
tracking_no
title
description
reporter_id
location_id
category_id
priority
photo_path
created_at
duplicate_of_id
duplicate_linked_by
duplicate_linked_at

IssueSupport
issue_id
resident_id
supported_at

Assignment
assignment_id
issue_id
worker_id
assigned_by
assigned_at
due_at
ended_at

ProgressUpdate
update_id
issue_id
updated_by
status_code
note
photo_path
created_at

2 List of Entities and Attributes

Underlined = primary key; *italic* = foreign key; *both styles* = both keys.
Multiple underlined attributes together form a composite primary key.

ResolutionReview
review_id
completion_update_id
reporter_id
decision
reason
reviewed_at

DeadlineChange
change_id
assignment_id
changed_by
old_due_at
new_due_at
reason
changed_at

Escalation
escalation_id
assignment_id
due_at_snapshot
raised_at
resolved_at
note

Comment
comment_id
issue_id
author_id
body
posted_at

Notification
notification_id
recipient_id
issue_id
message
created_at
read_at

RepairEstimate
estimate_id
issue_id
submitted_by
purpose
submitted_at

EstimateLine
estimate_id
line_no
description
quantity
unit_cost

EstimateDecision
estimate_id
reviewed_by
outcome
reason
decided_at

ActualExpense
expense_id
estimate_id
recorded_by
description
amount
spent_on
receipt_path
recorded_at

3 Relationships and Cardinality

Participation is written as (minimum..maximum); N means many. Optional 1:1 relationships allow zero or one related record. HAS_ROLE and IS_AFFECTED_BY map to AccountRole and IssueSupport.

Entity A / participation	Relationship	Entity B / participation	Type
Account (0..N)	HAS_ROLE	Role (0..N)	M:N
Account (0..N)	REPORTS reporter_id	Issue (1..1)	1:N
Category (0..N)	CLASSIFIES	Issue (1..1)	1:N
Location (0..N)	LOCATES	Issue (1..1)	1:N
Account (0..N)	IS_AFFECTED_BY	Issue (0..N)	M:N
Issue: duplicate (0..1)	IS_DUPLICATE_OF	Issue: canonical (0..N)	1:N
Account (0..N)	LINKS_DUPLICATE duplicate_linked_by	Issue (0..1)	1:N
Issue (0..N)	HAS_ASSIGNMENT	Assignment (1..1)	1:N
Account (0..N)	WORKS_ON worker_id	Assignment (1..1)	1:N
Account (0..N)	ALLOCATES assigned_by	Assignment (1..1)	1:N
Issue (1..N)	HAS_PROGRESS	ProgressUpdate (1..1)	1:N
Account (0..N)	POSTS_UPDATE updated_by	ProgressUpdate (1..1)	1:N
ProgressUpdate (0..1)	IS_REVIEWED_IN	ResolutionReview (1..1)	1:1
Account (0..N)	VERIFIES reporter_id	ResolutionReview (1..1)	1:N

3 Relationships and Cardinality

Participation is written as (minimum..maximum); N means many. Optional 1:1 relationships allow zero or one related record. HAS_ROLE and IS_AFFECTED_BY map to AccountRole and IssueSupport.

Entity A / participation	Relationship	Entity B / participation	Type
Assignment (0..N)	HAS_EXTENSION	DeadlineChange (1..1)	1:N
Account (0..N)	EXTENDS changed_by	DeadlineChange (1..1)	1:N
Assignment (0..N)	IS_ESCALATED_IN	Escalation (1..1)	1:N
Issue (0..N)	HAS_COMMENT	Comment (1..1)	1:N
Account (0..N)	WRITES author_id	Comment (1..1)	1:N
Account (0..N)	RECEIVES recipient_id	Notification (1..1)	1:N
Issue (0..N)	CONTEXT_FOR	Notification (0..1)	1:N
Issue (0..N)	HAS_ESTIMATE	RepairEstimate (1..1)	1:N
Account (0..N)	SUBMITS submitted_by	RepairEstimate (1..1)	1:N
RepairEstimate (1..N)	CONTAINS	EstimateLine (1..1)	1:N
RepairEstimate (0..1)	HAS_DECISION	EstimateDecision (1..1)	1:1
Account (0..N)	DECIDES reviewed_by	EstimateDecision (1..1)	1:N
RepairEstimate (0..N)	HAS_SPENDING	ActualExpense (1..1)	1:N
Account (0..N)	RECORDS_COST recorded_by	ActualExpense (1..1)	1:N

4 Foreign Keys

Each foreign key references the indicated primary-key column. All use ON DELETE RESTRICT and ON UPDATE RESTRICT. Nullable foreign keys represent optional references.

Foreign key	References	Nullable
AccountRole.user_id	Account.user_id	No
AccountRole.role_code	Role.role_code	No
Issue.reporter_id	Account.user_id	No
Issue.location_id	Location.location_id	No
Issue.category_id	Category.category_id	No
Issue.duplicate_of_id	Issue.issue_id	Yes
Issue.duplicate_linked_by	Account.user_id	Yes
IssueSupport.issue_id	Issue.issue_id	No
IssueSupport.resident_id	Account.user_id	No
Assignment.issue_id	Issue.issue_id	No
Assignment.worker_id	Account.user_id	No
Assignment.assigned_by	Account.user_id	No
ProgressUpdate.issue_id	Issue.issue_id	No
ProgressUpdate.updated_by	Account.user_id	No
ResolutionReview.completion_update_id	ProgressUpdate.update_id	No

4 Foreign Keys

Each foreign key references the indicated primary-key column. All use ON DELETE RESTRICT and ON UPDATE RESTRICT. Nullable foreign keys represent optional references.

Foreign key	References	Nullable
ResolutionReview.reporter_id	Account.user_id	No
DeadlineChange.assignment_id	Assignment.assignment_id	No
DeadlineChange.changed_by	Account.user_id	No
Escalation.assignment_id	Assignment.assignment_id	No
Comment.issue_id	Issue.issue_id	No
Comment.author_id	Account.user_id	No
Notification.recipient_id	Account.user_id	No
Notification.issue_id	Issue.issue_id	Yes
RepairEstimate.issue_id	Issue.issue_id	No
RepairEstimate.submitted_by	Account.user_id	No
EstimateLine.estimate_id	RepairEstimate.estimate_id	No
EstimateDecision.estimate_id	RepairEstimate.estimate_id	No
EstimateDecision.reviewed_by	Account.user_id	No
ActualExpense.estimate_id	RepairEstimate.estimate_id	No
ActualExpense.recorded_by	Account.user_id	No

5 Logical Schema

PK = primary key; FK -> Table.column = foreign key; UQ = unique; NULL = optional. Multiple [PK] attributes form one composite key.

Account(user_id [PK], full_name, phone [UQ], password_hash, house_flat [NULL], created_at, is_active)

Role(role_code [PK], role_name [UQ])

AccountRole(user_id [PK; FK -> Account.user_id], role_code [PK; FK -> Role.role_code], granted_at)

Location(location_id [PK], block_name, road_name, landmark [NULL])

Category(category_id [PK], category_name [UQ], description [NULL])

Issue(issue_id [PK], tracking_no [UQ], title, description, reporter_id [FK -> Account.user_id], location_id [FK -> Location.location_id], category_id [FK -> Category.category_id], priority, photo_path [NULL], created_at, duplicate_of_id [FK -> Issue.issue_id; NULL], duplicate_linked_by [FK -> Account.user_id; NULL], duplicate_linked_at [NULL])

IssueSupport(issue_id [PK; FK -> Issue.issue_id], resident_id [PK; FK -> Account.user_id], supported_at)

Assignment(assignment_id [PK], issue_id [FK -> Issue.issue_id], worker_id [FK -> Account.user_id], assigned_by [FK -> Account.user_id], assigned_at, due_at, ended_at [NULL])

ProgressUpdate(update_id [PK], issue_id [FK -> Issue.issue_id], updated_by [FK -> Account.user_id], status_code, note [NULL], photo_path [NULL], created_at)

5 Logical Schema

PK = primary key; FK -> Table.column = foreign key; UQ = unique; NULL = optional. Multiple [PK] attributes form one composite key.

ResolutionReview(review_id [PK], completion_update_id [FK -> ProgressUpdate.update_id; UQ], reporter_id [FK -> Account.user_id], decision, reason [NULL], reviewed_at)

DeadlineChange(change_id [PK], assignment_id [FK -> Assignment.assignment_id], changed_by [FK -> Account.user_id], old_due_at, new_due_at, reason, changed_at)

Escalation(escalation_id [PK], assignment_id [FK -> Assignment.assignment_id], due_at_snapshot, raised_at, resolved_at [NULL], note [NULL])

Comment(comment_id [PK], issue_id [FK -> Issue.issue_id], author_id [FK -> Account.user_id], body, posted_at)

Notification(notification_id [PK], recipient_id [FK -> Account.user_id], issue_id [FK -> Issue.issue_id; NULL], message, created_at, read_at [NULL])

RepairEstimate(estimate_id [PK], issue_id [FK -> Issue.issue_id], submitted_by [FK -> Account.user_id], purpose, submitted_at)

EstimateLine(estimate_id [PK; FK -> RepairEstimate.estimate_id], line_no [PK], description, quantity, unit_cost)

EstimateDecision(estimate_id [PK; FK -> RepairEstimate.estimate_id], reviewed_by [FK -> Account.user_id], outcome, reason [NULL], decided_at)

ActualExpense(expense_id [PK], estimate_id [FK -> RepairEstimate.estimate_id], recorded_by [FK -> Account.user_id], description, amount, spent_on, receipt_path [NULL], recorded_at)